

LOCAL LEAN VERIFICATION CERTIFICATE

Compact-Range Vietoris Powers of Countable Ordinals

Lucas Alves de Almeida | lucasdeirva8x@gmail.com

Verification date: 2026-09-28

Status: VERIFIED

The source files supplied with this certificate were compiled with Lean 4.19.0. The final audit checked the declared theorem types and printed their transitive axiom dependencies. The recorded audit exited successfully. This certificate records a local computation; it is not issued by an institution and does not assert independent peer review.

1. EXACT PRINCIPAL RESULT

For every mathlib ordinal a satisfying $\omega < a < \omega_1$, the compact-range Vietoris power of $\{b : \text{Ordinal} : b < a\}$ is second-countable and Lindelöf, but neither σ -compact nor Menger.

`VietorisOrdinals.ordinal_full`

The two ordinal inequalities are the only mathematical hypotheses of the principal theorem. Compactness, occurrence bounds, closedness, second countability and the conclusions are proved, not supplied as assumptions.

2. DEFINITIONS CHECKED

The carrier consists of maps from the natural numbers whose actual ranges are compact in the ground topology. The topology is induced from the topology generated by one open restriction on the whole range together with finitely many open coordinate restrictions. The ground ordinal subtype carries its order topology, inherited from the ordinary order topology on mathlib ordinals.

3. COMPILER, DEPENDENCIES AND TRUST

Lean: Lean (version 4.19.0, x86_64-w64-windows-gnu, commit 6caae842e94, Release)

mathlib revision: c44e0c8ee63ca166450922a373c7409c5d26b00b

Allowed and reported transitive axioms: `propext`; `Classical.choice`; `Quot.sound`. The recorded final declarations have no dependence on `sorryAx`, custom mathematical axioms, or native-computation trust axioms. This relies on the standard Lean kernel, compiler installation, and pinned mathlib library.

4. AUDITED DECLARATIONS AND SCOPE

VietorisOrdinals.second_countable

The compact-range Vietoris power with countable index is second-countable for every second-countable ground space.

VietorisOrdinals.not_sigmaCompact

An injective enumeration of a compact convergent-sequence-type set with isolated finite values yields non-sigma-compactness, by the delayed diagonal.

VietorisOrdinals.not_menger

Under the same structural hypotheses, the compact-range power fails the standard Menger selection property for arbitrary open covers.

VietorisOrdinals.ordinal_main

For every actual ordinal between omega and omega one, the specified compact-range power is second-countable, Lindelof and not sigma-compact.

VietorisOrdinals.ordinal_full

The complete principal ordinal theorem, including failure of the Menger property.

VietorisOrdinals.ordinal_subsets_countable

Every subset of each countable ordinal interval has cardinality at most aleph zero.

The paper's classification for uncountable ordinals is attributed to Caruvana and Holshouser; this certificate does not claim a new formalization of their Proposition 2.17. The closed-subspace transfer proposition is given with a mathematical proof in the paper. The source-PDF interpretation and the cardinal-index convention are documented separately from the kernel-checked topology and ordinal statements.

5. REPRODUCTION AND INTEGRITY

The source package includes lean-toolchain, lakefile.lean, lake-manifest.json, the Lean modules, FullAudit.lean, verify.ps1, the complete successful audit log, and a machine-readable verification-status.json with SHA-256 hashes. A fresh machine must first install the pinned Lean version and fetch the pinned mathlib dependencies. Then run verify.ps1 from the project directory. The script stops on any failed compilation and rejects unexpected axioms.

Audit log SHA-256:

1cefae48b628a40bdd34c5e736dad253 2d188789511106037610d22eefe62507

Source manifest SHA-256:

c09a1406ef6a2e1d24d68eba05f06f21 926f51a66c3ee345a814844a3f6be828

Hashes identify the verified source and log bytes; they do not replace rerunning Lean. The dependencies and toolchain are not embedded in the source archive. All compiler output and any scope limitations are preserved in the package.

6. ATTRIBUTION

Lucas Alves de Almeida proposed and guided the generalization of his earlier convergent-sequence result. ChatGPT-Astra developed the main general proof and prepared the formalization, checks, manuscript, and this local verification record under his direction.